

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018-2019 (Even Semester)

Innovative Practices Description

UNIT II

Degree, Semester & Branch : V Semester B.E. EEE
Course Code & Title : EE8401 Electrical Machines- II
Name of the Faculty member : Mr. E. Thangam, AP (SG)-EEE

Name of the Topic : Synchronous machine, Induction Machine Operations
Name of the Innovative Practice : Demonstration Classes with Demo Machine
Date & Duration : 22.03.2018
ICT Tool Used : -

Description:

I taught Electrical Machines-II course with demo DC machines and Transformers available in Electrical Machines laboratory. I asked the students to troubleshoot the machine problems, and then causes and effects were analyzed. I conduct classes to fill up gap between the theory and practical.

We have demo generator, practical transformer, machine core stampings structure, varnish coated windings etc. This way of teaching is very much helpful for the student's better understanding.



A Half day field visit to Power House at Ramco Institute of Technology, Rajapalayam on 22.03.2018 from 10.50 AM to 12.35 PM for the II year students of EEE. The students were accompanied by two faculty members, Mr.E.Thangam, AP/EEE, Mr.D.KarthikPrabhu, The students were formed in batches and were explained the different components in a Power House such as transformer, bus bars, circuit breakers and generators. Also the students were taken to the transformer (1250 kVA) yard, where innovate lectures of EE6401 Electrical Machines-I on the topic of "Transformer principle of operation" was conducted and students were made to clarify their doubts. The visit came to an end at 12.35 PM.



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A Half day field visit to Power House at Ramco Institute of Technology, Rajapalayam on 24.03.2017 from 2.30 PM to 5.00 PM for the II year students of EEE. The students were accompanied by three faculty members, Mr.E.Thangam, AP/EEE, Mr.D.KarthikPrabhu, Ms.M.Eswari AP/EEE. The students were formed in batches and were explained the different components in a Power House such as transformer, bus bars, circuit breakers and generators. Also the students were taken to the transformer (1250 kVA) yard, where innovate lectures of EE6401 Electrical Machines-I on the topic of “Transformer principle of operation” was conducted and students were made to clarify their doubts. The visit came to an end at 5.00 PM.



Goals (Learning Outcomes):

- ❖ The students will be able to explain the construction and operation of transformer, DC machines, AC machines
- ❖ The students will be able to compare operating principle of DC machines and AC machines.

Use of appropriate methods:

Justification for choosing the topic using demonstration and field visit activity:

When compared to PPT presentation/ Black board presentation/ animated videos, the demonstration and field visit classes are very effective to teach construction and operation of various electrical machines by demo machine or real time machines. After teaching the concept, the students will be able to explain the construction and operation of various electrical machines. This would be useful for us to judge the understanding level of the students.

Time Allotted for the Activity: 50 minutes

Effective presentation

Total Strength is 56

Reporter: Myself

Time Allotted for the Activity: 50 minutes

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, students were shown more interest to troubleshoot the operation of various electrical machine operation.

- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 80% of students answered correctly.**

Reflective Critique:

• **Benefits:**

- Since I had handled the subject for many times usually the students will confuse with the principle of operation of various machines.
- Time utilization for conducting activity.

• **Challenges:**

- Since the classes were conducted in RIT power house and laboratory, electrical safety is important. Handling 55 students in one area with HT supply is challenging one.

Changes required for future (if required): Nil

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	3	2		2	2		1		3
CO2	3	1	2	2	2		1	1		1		3
CO3	3	1	1	1	3		1	1		1		3

CO1: Explain the concept, principle and performance of salient and non salient type synchronous generator.

CO2: Explore the concept, principle and performance of synchronous motor.

CO3: Illustrate the Construction, working principle and performance of three phase induction motor.

References:

1. A.E. Fitzgerald, Charles Kingsley, Stephen. D. Umans, 'Electric Machinery', Mc Graw Hill publishing Company Ltd, 2003.
2. Vincent Del Toro, 'Basic Electric Machines' Pearson India Education, 2016.
3. Stephen J. Chapman, 'Electric Machinery Fundamentals' 4th edition, McGraw Hill Education Pvt. Ltd, 2010.